

BUILDING AN ARTIFICIAL INTELLIGENCE CHATBOT FOR IT SUPPORT AT CUT

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ABSTRACT

Since their debut in the 1960s, chatbots have proven to be a force to be reckoned with in the IT space. This technology has gained a well-deserved popularity for its efficiency, convenience and, most importantly, improved user satisfaction in the industries that it has been implemented in.

This research aims to outline the potential benefits of implementing an AI chatbot for IT support at CUT. The research starts by providing insight into the existing system for IT support at CUT. Then, it follows by outlining the major benefits of an AI chatbot, which are: 24/7 availability, immediacy and automation, and how integrating an AI chatbot into the existing system would enhance the existing system.

This will be done by collecting requirements by means of use-case scenarios, surveys and observation of end users. These findings will be used for decision-making in the subsequent stages of the SDLC. For the design phase, tools like diagrams.net, and Chatbase will be used for the working solution prototype. And finally, for the development phase, the tools that will be used are HTML, CSS and JavaScript for creating a mock-up CUT website where the chatbot will be embedded, while Voiceflow will be used for building the chatbot. Proper testing on the same platform (Voiceflow) as well as beta testing will be prioritized to ensure good quality and a product that meets the user requirements. Overall, this research aims to hopefully contribute to AI-driven solutions at CUT.

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LITERATURE REVIEW

INTRODUCTION

A chatbot is software that interacts with humans using speech, text or both. Chatbots were designed to imitate the role of a human agent in specific cases (Casas et al., 2020). They are designed to be able to understand and respond to conversations in a human-like manner, giving the impression of being human. The technology was first introduced in 1966 by Joseph Weizenbaum when he introduced a chatbot named Eliza (Casas et al., 2020). This technology has since been evolving and gaining popularity across all social media.

This literature review will investigate artificial intelligence and its usefulness in developing a chatbot system for IT support at CUT. By the utilization of a chatbot system, CUT can have a 24/7 and immediate support response. The questions that will be dealt with are: What are the challenges confronting the existing IT support mechanisms at CUT? To what extent will the chatbot effectively resolve IT support queries and issues compared to human support agents? What is the feasibility of integrating the AI chatbot with the existing IT infrastructure at CUT? By reviewing existing literature, this study will clarify these questions by providing insights into the potential benefits of deploying an AI chatbot system for IT support at CUT. By utilizing findings from diverse sources including academic research, case studies and journals, this review aims to contribute to the discourse surrounding AI-driven innovations in educational settings.

Furthermore, this literature review aims to identify gaps and areas for further research within the existing scope. In the following sections, we will look at the thematic applications of chatbot systems.

THEMATIC SECTIONS

IT SUPPORT

Information technology (IT) support refers to the services and help offered to users inside an organization to guarantee the proper operation, maintenance, and troubleshooting of IT systems and technological infrastructure. It includes a variety of tasks such as troubleshooting technical difficulties, aiding users with software and hardware problems, managing network setups, implementing cybersecurity measures, and maintaining IT assets. All organizations rely on the efficiency of IT support to run well.

At present, within the context of the institution, Central University of Technology (CUT), the scope of IT support encompasses tasks such as password management, including both resets and initial creations, for online portals utilized by students and staff members alike. Additionally, it involves the installation of software applications and troubleshooting of Wi-Fi connectivity issues. The current mechanism employed by CUT for managing IT support matters is through the utilization of an IT support service desk, which operates on a ticket-based system. Users are required to log their IT issues as tickets and await the attention of an IT support agent to address their concerns. This existing framework exhibits a notable deficiency in immediacy, thereby necessitating the exploration of alternative solutions, such as the implementation of a chatbot system. An additional deficiency with this system is the considerable time spent by the IT support agents in attending to repetitive queries and issues. Implementation of a chatbot system could alleviate this challenge by efficiently handling frequently encountered queries, resolving repetitive issues and escalating matters beyond its scope to the human support agents.

CHATBOT

According to (Casas et al., 2020), "Chatbots are computer programs aiming to replicate human conversational abilities through voice exchanges, text dialogues or both."

Chatbots have been gaining popularity and evolving since the first time they were introduced to us in by Joseph Weizenbaum in 1966 (Nguyen, n.d.). At present, chatbots are gaining the most popularity in certain industries like customer support, e-commerce as well as education.

CHATBOTS IN THE CONTEXT OF EDUCATIONAL SETTINGS

Research shows that there is a need for chatbots in the context of educational settings, specifically for student support. Secondly, it has also been proven that chatbot integration in educational settings has been successful (Dang Truong Ho Chi et al., 2020). " It is easier, both mentally and practically, for a student to answer an SMS compared to initiating contact with a support professional, says (Pesonen, 2021). In the context of using chatbots for customer support, that has also been proven to have a high success rate as well as high user satisfaction rate (Fiore et al., 2019).

CONCLUSION

Using the above research findings, this literature review concludes that there are potential benefits in implementing an AI chatbot system for IT support at CUT. The existing IT support mechanisms face challenges in terms of immediacy and efficiency, which can be addressed by a chatbot system. There is evidence of increased user experience and operational efficiency in organizations after the implementation of a chatbot system. By utilizing this AI technology, CUT can also benefit from those advantages. Overall, this review highlights the potential benefits of exploring the integration of AI chatbots with the existing IT infrastructure.

PROBLEM STATEMENT

The problem statement that this research proposal states is that the current CUT IT support system is lacking in immediacy and efficiency. This is caused by the system being reliant on human support assistants, which results in IT support services at CUT lacking 24/7 availability and immediacy. Another cause of these challenges is the time spent by IT support assistants attending to repetitive trivial queries, thus causing delay in the resolution of high-end queries. The solution to this problem statement is an AI chatbot that is trained to resolve trivial IT support queries and escalate the ones that are not in its scope. This integration of an AI chatbot in the CUT IT support system will not only enhance the efficiency and immediacy of the system but it also has potential to improve user satisfaction as this technology is renowned for doing so.

RESEARCH QUESTIONS, AIM AND OBJECTIVES

Research questions

- What are the current trends in the integration of artificial intelligence chatbots in the education sector and other various sectors?
- How can an artificial intelligence chatbot enhance the operation of IT support at CUT?
- What are the main challenges in the development and implementation of an AI chatbot for IT support at CUT?

Research aim

The aim of this study is to investigate the current trends in AI chatbot integration in various sectors or industries but with a focus on the education sector. By doing this, this research aims to contribute to the advancement of AI-driven solutions to enhance operations at CUT.

Research objectives

1. *To review current literature and technological advancements in AI chatbots for the enhancement of operations in various sectors.*

This objective will be focused on conducting a literature review to identify the latest trends, developments, challenges, opportunities in the implementation of AI chatbots for IT support and other operations in other industries.

2. *To evaluate the effectiveness of AI chatbots in the enhancement of IT support services at CUT.*
3. *To build an AI chatbot for IT support at CUT.*

This objective entails designing a prototype and building an AI chatbot that is going to be trained to deal with the trivial IT support issues at CUT. The system will be tested thoroughly for the accuracy of executing tasks.

RESEARCH METHODOLOGY

In this research project, for data collection and usage, a quantitative approach will be used. Online surveys using Microsoft Forms will be used to determine students' needs and expectations. For the development of the solution, a waterfall methodology will be undertaken. Waterfall methodology is a project management approach that follows a sequential and linear process. It is made up of a total of 5-6 phases and each phase does not commence before the completion of the previous phase. The breakdown of the phases of this project will be as follows: Requirements gathering and analysis phase, Design phase, Development phase, Testing phase and Deployment.

Requirements gathering and analysis phase

This is where online surveys will be sent out to end users (students) to determine their needs and expectations in the context of IT support.

Methods

Use case scenarios: Use case scenarios will be used to visualize real-life interaction between the chatbot and the user to refine user interactions and foresee challenges that may be anticipated.

Online surveys: This method will be used to collect data from users for the purpose of getting more insights on user needs.

Observations: This method will be used to gain more insights of how the users interact with the current IT support system, filling the gaps that the previous methods could not fill.

Design phase

Tools

Diagrams.net chosen because of its intuitive user interface that is easy to navigate, will be used to design use-case diagrams.

Chatbase: Chosen for its ability to provide design iterations based on the relevant data.

MS Word: Chosen for its ability to produce high quality WBS and Product Breakdown Structures seamlessly.

MS Excel: Chosen for the purpose of creating the GANNT chart with minimal effort without compromising the quality of the output.

Development, Testing and deployment phases

Tools

HTMLS, CSS and JavaScript: will be used for developing a CUT website mock-up where the AI chatbot will be embedded.

Voiceflow: This tool has been selected because of its ability to build, test and deploy chatbots seamlessly.

WORK BREAKDOWN STRUCTURE

WBS code	Task Name	Description
1.0	Project Initiation	Initial phase to define the project and its objectives.
1.1	Develop WBS, PBS and PMP	Outline the tasks and products required for the project.
2.0	Planning	Planning phase to gather requirements and develop project plans.
2.1	Requirement gathering	Collect requirements through use-case scenarios, surveys and observation.
2.1.1	Use-Case Scenarios	Develop use-case scenarios to outline the chatbot's functionality.
2.1.2	Surveys and Observations	Gather user input and observe the current IT support processes.
3.0	Design	Design phase to create system architecture and prototypes.
3.1	Create system architecture	Design the system's architecture including workflows and data flows.
3.1.1	Design workflow diagrams(diagrams.net)	Create detailed diagrams to visualize the project's processes.
3.1.2	Prototyping	Design the user interface and user experience for the chatbot integration.
3.2	Review and approve designs	Conduct design reviews and get supervisor's approval.
4.0	Development	Development phase to build and test the chatbot and website.
4.1	Frontend Development	Develop the frontend interface of the CUT website.
4.1.1	Develop mock-up CUT website (HTML, CSS, JavaScript)	Create a mock-up of the CUT website using HTML, CSS and JavaScript.
4.2	Chatbot Development	Develop the AI chatbot and integrate it into the website.

4.2.1	Build Chatbot (Voiceflow)	Use Voiceflow to create the chatbot's logic and interaction flow.
5.0	Testing	Testing phase to ensure the chatbot meets requirements and functions well.
5.1	Functional testing (Voiceflow)	Testing the chatbot's functionality on Voiceflow
5.2	Beta Testing with End Users	Conduct beta testing with selected end-users to gather feedback.
5.3	Review and fix issues	Address any issues or bugs found during testing.
6.0	Implementation	Implementation phase to deploy the chatbot and monitor its performance.
6.1	Deploy chatbot on CUT website	Deploy the chatbot on the CUT website.
6.2	Monitor performance and user feedback	Continuously monitor the chatbot's performance and gather user feedback.
7.0	Project Closure	Final phase to close the project.
7.1	Documentation	Complete all project documentation.
7.2	Project review and lessons learned.	Review the project and document lessons learned for future reference.
7.3	Close project	Formal closure of the project including final approvals.

PROJECT MANAGEMENT PLAN

Project Objectives:

1. To analyze the current IT support system at CUT.
2. To identify potential benefits and improvements through the integration of an AI chatbot.
3. To design and develop an AI chatbot using Voiceflow.
4. To test and validate the chatbot to ensure it meets user requirements.

Scope:

- **Requirements Collection:** Use-case scenarios, surveys, and user observations.
- **Design Phase:** Tools like diagrams.net and Figma for showcasing designs.
- **Development Phase:** HTML and CSS for creating a mock-up CUT website, and Voiceflow for building the chatbot.
- **Testing Phase:** Proper testing on Voiceflow and beta testing to ensure quality.

Deliverables:

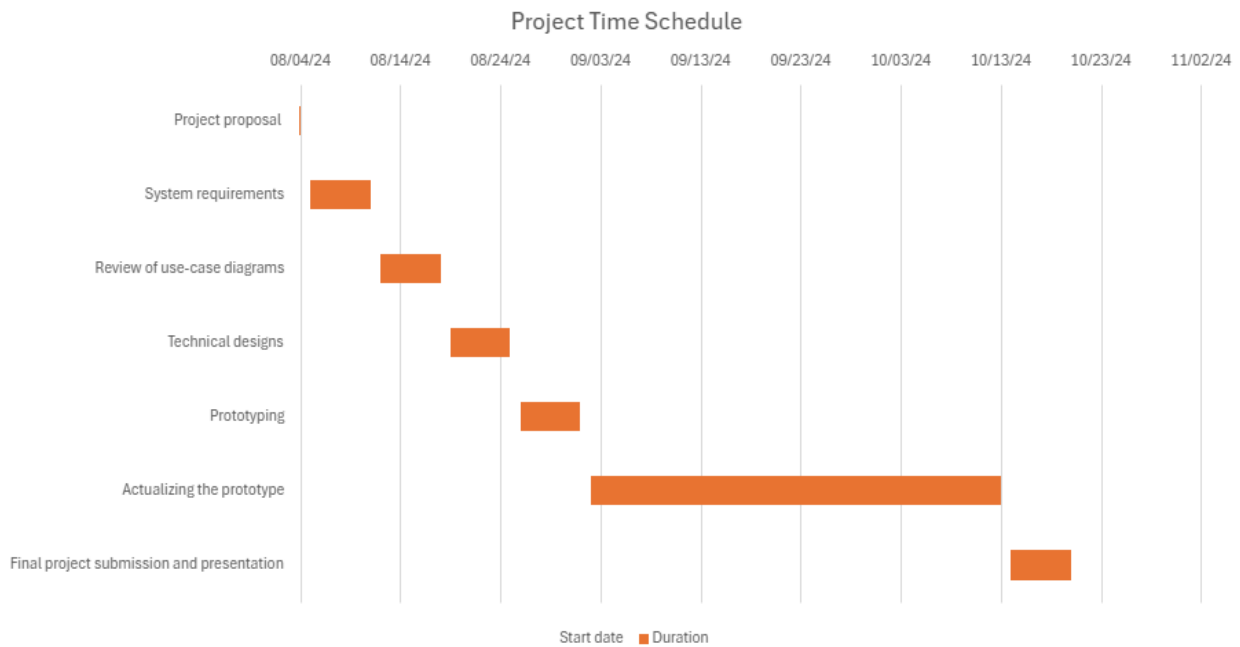
1. Detailed report on the current IT support system and potential improvements with an AI chatbot.
2. Design prototypes using diagrams.net and Figma.
3. A functional mock-up CUT website with the embedded chatbot.
4. A fully developed and tested AI chatbot ready for implementation.

Responsibilities:

- **Yonela Gingqi:** Conducting research, collecting requirements, designing, developing, and testing the AI chatbot.

MILESTONES AND PROJECT SCHEDULE

Project proposal	07/29/24 - 8/4/2024
System requirements	08/05/24 - 08/11/2024
Review of use-case diagrams	08/12/24- 08/18/2024
Technical designs	08/19/24 - 08/25/2024
Prototyping	08/26/24 - 09/01/2024
Actualizing the prototype	09/02/24 - 10/13/2024
Final project submission and presentation	11/01/2024



SYSTEM ARCHITECTURE

Three-tier client-server architecture

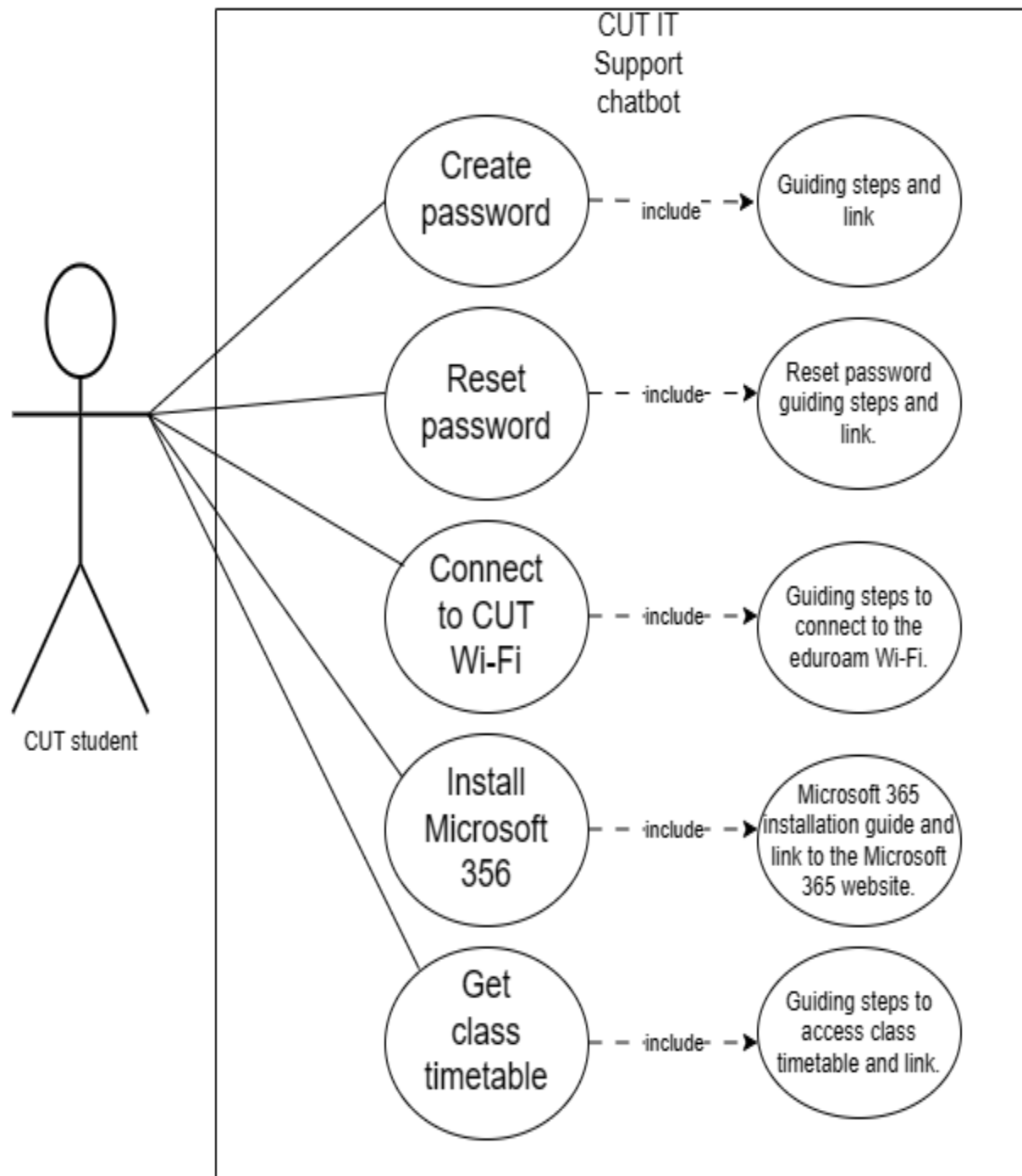
The three-tier architecture is a client-server architecture that encompasses a presentation layer, an application layer, and a data layer. This system architecture was chosen because of its suitability for the project hand. This is how the 3-tiers of this project are layered:

Presentation layer (HTML, CSS, JavaScript, Voiceflow): The website was developed using HTML, CSS, and JavaScript, while the GUI of the chatbot was built in Voiceflow. The presentation layer communicates with the other layers through API calls. The chatbot is embedded on the website using JavaScript.

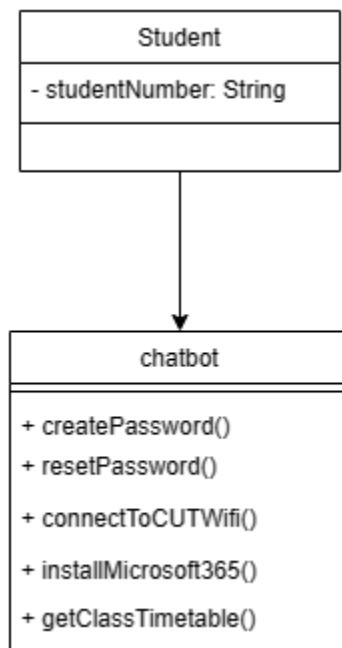
Application layer (Voiceflow): This layer (managed in Voiceflow) manages intent recognition and dialogue management.

Data layer (Voiceflow): stores chat knowledge base, and workflows that assist the chatbot in generating accurate responses.

USE CASE DIAGRAM



CLASS DIAGRAM

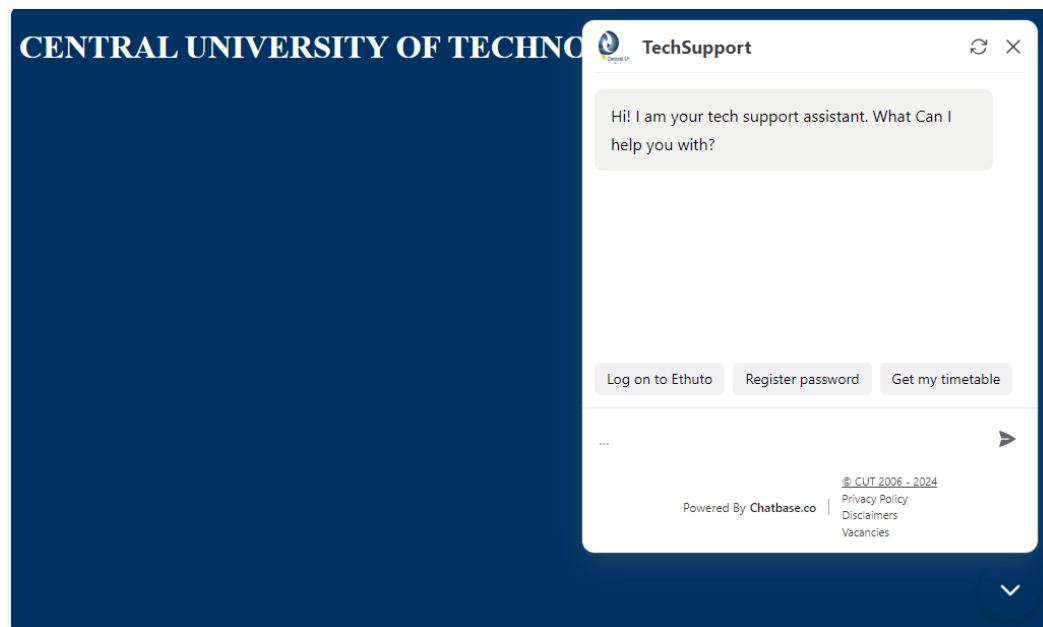


SYSTEM PROTOTYPE

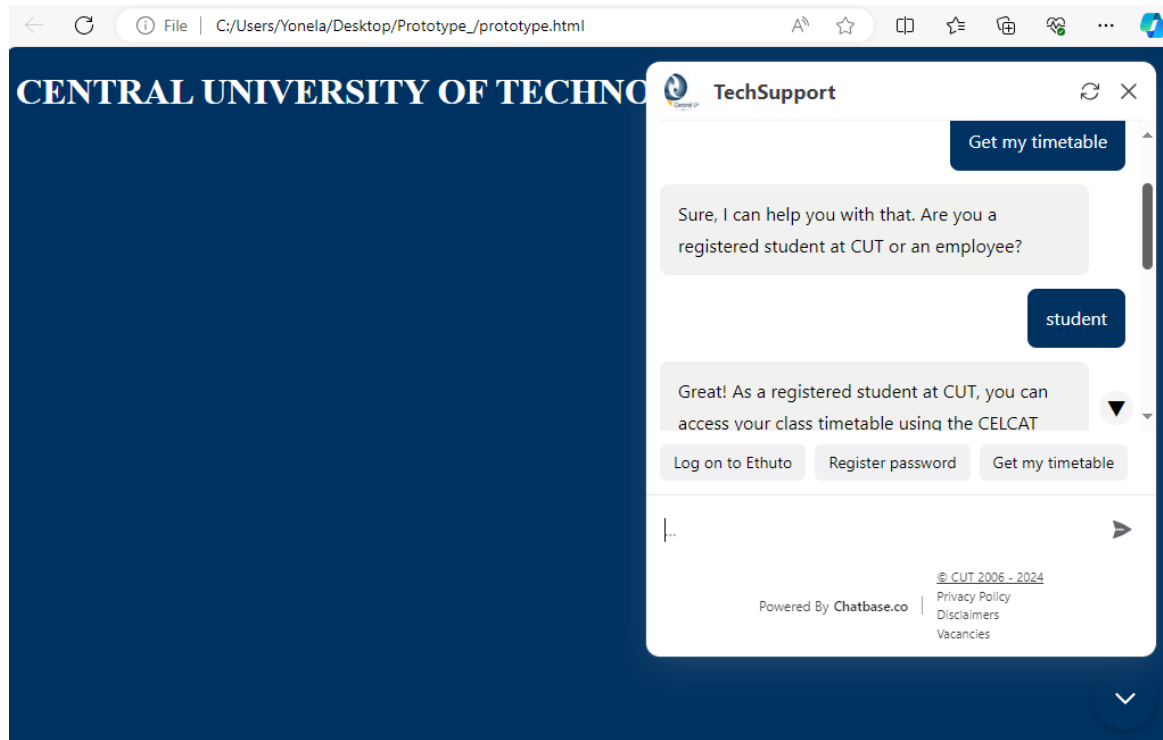
A demonstration of the chatbot on a mock-up CUT website. It sits at the bottom of the page in the form of a bubble.



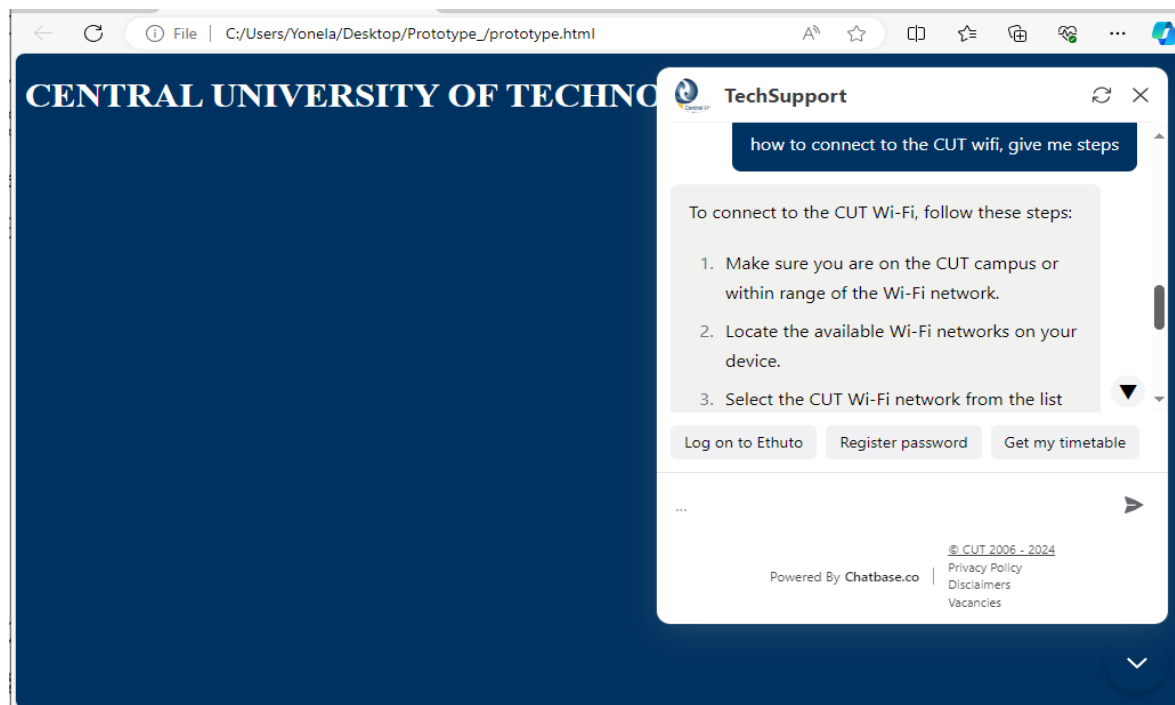
When the user clicks the button to chat with the chatbot:



The chatbot will be able to ask follow-up questions to gain more clarity



Replies will be in the form of steps or guidelines:





Microsoft 365

If you need assistance with Microsoft 365, here are some steps and resources that might help you:

Setting Up Microsoft 365:

1. Register Your Password:

- New students should first register their password at [this link](#).

Register password

Reset password

Get my timetable

|...



Additional Resources:

- **How to Enroll and Reset Password:**
 - Watch the how-to video [here](#).

If you encounter any issues or need further assistance, you can contact IT support:

- **Bloemfontein Campus:**
 - Phone: 051 507 3059
 - Email: smjoubert@cut.ac.za

Register password

Reset password

Get my timetable

|...



The user will be able to ask to be directed to IT support to be helped by a system administrator or IT support technician.

 **TechSupport** 

I want to talk to IT support

Sure, I can help you with that! Here are the contact details for IT support at CUT:

IT Support Contact Information:

- Bloemfontein Campus:
 - Phone: 051 507 3059
 - Email: smjoubert@cut.ac.za
- Welkom Campus:

Register password

Reset password

Get my timetable

When the user asks a question not related to the user case scenarios, the chatbot will be able to reply appropriately.

 **TechSupport** 

Hi! What can I help you with?

where is kfc?

I'm sorry, but I can only assist with information related to the Central University of Technology (CUT). If you have any questions about CUT, feel free to ask!

[Register password](#) [Reset password](#) [Get my timetable](#)



CONCLUSION

Achievements

- Development of a CUT mock-up website using HTML, CSS and JavaScript.
- Building an AI chatbot using Voiceflow.
- Training of the chatbot on Voiceflow.
 - The chatbot can assist the user with trivial IT support queries.
 - It can also assist the user with contacting the IT support department should the user be unsatisfied with the assistance of the AI agent.
 - The chatbot successfully uses the knowledge base and workflows to accurately reply to queries.
- Testing: the chatbot was successfully tested on Voiceflow before deployment.
- The chatbot was successfully embedded on the CUT mock-up website.

Recommendations

There are gaps that can be filled to improve the quality of this work. The areas of improvement from my point of view are as follows:

- Developing the chatbot from scratch using the Python programming language instead of Voiceflow. Python supports advanced usage of NLP (Natural language Processing) and machine learning. Using these python abilities would drastically improve the capabilities of the chatbot in various ways, two of them being: allowing the chatbot to handle multiple requests at once and getting better at responses by learning from the user's questions. These improvements would improve the quality of the responses of the chatbot, ultimately leading to increased user satisfaction.
- Getting the AI agent to directly escalate issues by lodging tickets on behalf of the user instead of providing the user with information on how to lodge a ticket or providing them with an email to contact IT support. Taking this weight off the user's shoulders would have a positive impact on user satisfaction.
- Expanding the scope of the chatbot to be more helpful by adding more objectives. Two ways of doing this would be assisting the CUT community with more than just IT support but not go beyond the CUT scope and gathering input straight from the

end-users on how they would like the scope of the chatbot to be expanded so that it may be more of assistance to them.

With the above-mentioned improvements, this work would not only be of good quality and great assistance to the CUT community, but it would also serve as a solid foundation or introduction to the world of artificial intelligence and machine learning for the researcher/developer.

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